## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Values and other decisional factors regarding treatment of hypercalcemia of malignancy: A Systematic Review Protocol |
| **AUTHORS** | Bassatne, Aya; Rahme, Maya; Piggott, Thomas; Murad, M. Hassan; Hneiny, Layal; El-Hajj Fuleihan, Ghada |

## VERSION 1 – REVIEW

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| **REVIEWER** | Brun, Vegard Heimly<br>University Hospital of North Norway |
| **REVIEW RETURNED** | 31-Mar-2021 |

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| **GENERAL COMMENTS** | Minor points:<br>Introduction:Parathyroid carcinomas are usually treated with surgery, which is not mentioned. I do understand that only medical therapy is of interest to the study, but still surgery should be mentioned as a treatment option in this section.<br><br>Objectives<br>First sentence is hard to read, not sure what you mean by “other such factors”. I think this paragraph says nothing about the study’s objective, and belongs to another section, for example Introduction. Objectives are described better in the Abstract.<br><br>Data collection<br>I think contacting authors to obtain unpublished information should be avoided, as the evaluations should be based on the published available information.<br><br>Data Synthesis<br>“Two independent reviewers will identify themes from each article until saturation and reach consensus on how the themes would converge into unique.”<br>Please explain what this means. |

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| **REVIEWER** | Chukir, Tariq<br>Weill Cornell Medicine - Qatar |
| **REVIEW RETURNED** | 12-Apr-2021 |

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| **GENERAL COMMENTS** | Decisional and contextual factors play an important role in the shared decision making process. While these factors are considered daily in clinical practice, there is limited data on this topic in the literature.<br><br>My concern with this study is that the treatment options available for HOM are limited. |

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| | I disagree with the researchers that we have many treatments for HCM to choose from. When a patient with HCM meets an indication for pharmacotherapy, the firstline treatment is a bisphoshonate, ideally zoledronic acid. Denosumab could be considered in patients with refractory HCM or renal disease. Glucocorticoids have certain indications in a subset of patients. Calcitonin could be used as a bridge but not effective long-term. . The role of cinacalcet is unclear in PTH-independent hypercalcemia.<br><br>Decisional factors e.g. patient's values, costs, resources, feasiblity, acceptability and equity are important to consider. However, I'm worried about the validity and generalizability of the results. It is important to keep in mind when interpreting the literature that these decisional factors may have changed over the years after the study had been published (for example, cost and accessability of the medication). Also, these factors are population dependent and may not be generalized to other populations. |

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| **REVIEWER** | Angelousi, Anna<br>National and Kapodistrian University of Athens |
| **REVIEW RETURNED** | 29-May-2021 |

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| **GENERAL COMMENTS** | The authors present analytically and extensively all the steps required for a methodologically correct systematic review.<br>I was wondering if the authors intend to proceed to a metanalysis (for exemple it would be probably interesting to show if possible the cut-off of calcium levels for some specific therapeutic decision, also if mortality or OS or PFS are associated with calcium levels or treatments?) If so, then statistical analysis should be incorporated.<br>Furtermore it is recommended that study is registrated in PROSPERO. |

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| **REVIEWER** | O'Sullivan, Dylan<br>University of Calgary, Oncology |
| **REVIEW RETURNED** | 02-Jun-2021 |

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| **GENERAL COMMENTS** | Abstract<br>- Saying subgroup analyses seems incorrect for this type of review. I suggest changing to: "with possible subgroup summaries by drug class or etiology of hypercalcemia of malignancy"<br>- Please change "will assist" on line 40 to "may assist", since it depends on the quantity and quality of the literature<br>- I suggest changing scarcity to "potential scarcity" on line 43, since this is unknown at this point.<br><br>Introduction<br>- Sentence 3 would be easier to follow if the mechanisms were numbered<br>- Line 25 – is this statistic up-to-date? It seems like this statistic comes from a small study (126 patients) published in 1990 and not reference 7. Reference 8 talks about a mortality rate in hospital of 6.8%. Please ensure accurate and up-to-date survival outcome estimates are cited.<br>- RANKL needs to be spelled out – page 3, line 38<br>- GRADE should be spelled out here – page 3, line 55<br><br>Methods |

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| | - What is the rationale for limiting to the last 10 years? Please include those reasons in the protocol.<br>- Why was there a search also done with no time limit and only in MEDLINE?<br>- Consider adding percent agreement and kappa statistics for agreement between reviewers in each stage.<br>- This review may benefit from also screening any studies that have cited your included studies – as part of the hand search. Please consider including in the protocol.<br>- For eligibility – please specify the types of observational studies included.<br>- Abbreviations should be footnoted in the tables |

## VERSION 1 – AUTHOR RESPONSE

Reviewer 1: Dr. Vegard Heimly Brun, University Hospital of North Norway

Introduction: Parathyroid carcinomas are usually treated with surgery, which is not mentioned. I do understand that only medical therapy is of interest to the study, but still surgery should be mentioned as a treatment option in this section.
We acknowledge the reviewer's comment and have not excluded surgical interventions from our review. However, this systematic review aims to inform guidelines of the Endocrine Society focusing on the medical treatment of hypercalcemia of malignancy.

Objectives: First sentence is hard to read, not sure what you mean by "other such factors". I think this paragraph says nothing about the study's objective, and belongs to another section, for example Introduction. Objectives are described better in the Abstract.
Thank you for your comment. We have modified the Objectives section as follows: "The objective of this systematic review is to summarize the available evidence on important decisional factors including physicians' and patients' values and preferences, cost, acceptability, feasibility, and equity, for the development of guidelines for the treatment of hypercalcemia of malignancy."
We have also moved the following sentence to the Introduction: "The evidence to decision framework from the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) Working Group (Evidence to Decision, EtD) describes five other such factors: patient's values, costs and resources, feasibility, acceptability, and equity [22]. Therefore, and to better inform the recommendations to be made by the Endocrine Society in its Clinical Practice Guidelines on Treatment of Hypercalcemia of Malignancy, we decided to conduct a rigorous meta-narrative systematic review to summarize the best available evidence about the above described decisional factors."
We have moved and modified the following sentence in the Methods: "We initially used a search strategy through Epistemonikos database to identify any prior systematic review that addressed factors related to decisional frameworks (Supplemental material 1A), in the treatment of hypercalcemia of malignancy [24]. We were unable to find any relevant publication on the topic (Supplemental material 1B)."

Data collection: I think contacting authors to obtain unpublished information should be avoided, as the evaluations should be based on the published available information.
Thank you for your comment, we have deleted this section from our Methods.

Data Synthesis: "Two independent reviewers will identify themes from each article until saturation and reach consensus on how the themes would converge into unique."
Please explain what this means.

Thank you for your comment. We have modified this section to clarify the process as follows: “Two independent reviewers will identify themes from each article that can map to a concept in the evidence to decision framework. For example, a theme about whether patients prefer a certain treatment characteristic can map to the acceptability domain in the framework. Consensus among the two reviewers about themes is reached via discussion. We will seek a state of saturation in which the two reviewers are reasonably assured that further data collection would yield similar results. The next step after saturation is to confirm emerging themes and conclusions.”

Reviewer 2: Dr. Tariq Chukir, Weill Cornell Medicine – Qatar

Decisional and contextual factors play an important role in the shared decision making process. While these factors are considered daily in clinical practice, there is limited data on this topic in the literature. My concern with this study is that the treatment options available for HOM are limited.
I disagree with the researchers that we have many treatments for HCM to choose from. When a patient with HCM meets an indication for pharmacotherapy, the firstline treatment is a bisphoshonate, ideally zoledronic acid. Denosumab could be considered in patients with refractory HCM or renal disease. Glucocorticoids have certain indications in a subset of patients. Calcitonin could be used as a bridge but not effective long-term. . The role of cinacalcet is unclear in PTH-independent hypercalcemia.
Decisional factors e.g. patient’s values, costs, resources, feasiblity, acceptability and equity are important to consider. However, I’m worried about the validity and generalizability of the results. It is important to keep in mind when interpreting the literature that these decisional factors may have changed over the years after the study had been published (for example, cost and accessability of the medication). Also, these factors are population dependent and may not be generalized to other populations.
We thank the reviewer for his insightful comment. We have modified the term “many” and replaced it with a more appropriate term “several”. Although IV Zoledronate may be the first line therapy used in a tertiary care setting, it may not be the most accessible and feasible in other settings. We have added to the Strengths and Limitations section following the Abstract the fact that some of our findings might be outdated with regards to cost and not generalizable across different populations. Importantly, we are unaware of any systematic review investigating decisional frameworks that are instrumental in informing the development of clinical practice guidelines on the treatment of hypercalcemia of malignancy.

Reviewer 3: Dr. Anna Angelousi, National and Kapodistrian University of Athens
The authors present analytically and extensively all the steps required for a methodologically correct systematic review. I was wondering if the authors intend to proceed to a metanalysis (for exemple it would be probably interesting to show if possible the cut-off of calcium levels for some specific therapeutic decision, also if mortality or OS or PFS are associated with calcium levels or treatments?) If so, then statistical analysis should be incorporated.
We thank the reviewer for these interesting comments. However, we have decided to pursue a narrative approach and therefore we prefer not to conduct any meta-analyses.

Furtermore it is recommended that study is registrated in PROSPERO.
Thank you for your comment, we have submitted the protocol to PROSPERO: CRD42021264371.

Reviewer4: Dr. Dylan O'Sullivan, University of Calgary

Abstract:
Saying subgroup analyses seems incorrect for this type of review. I suggest changing to: “with possible subgroup summaries by drug class or etiology of hypercalcemia of malignancy”
Thank you for your comment, we have modified as per your suggestion: “The outcomes of interest are

the decisional factors that influence drug selection, with possible subgroup summaries by drug class or etiology of hypercalcemia of malignancy."

Please change "will assist" on line 40 to "may assist", since it depends on the quantity and quality of the literature
This sentence has been deleted as per the editor's recommendations.

I suggest changing scarcity to "potential scarcity" on line 43, since this is unknown at this point.
We have modified the sentence as per your suggestion: "The potential scarcity of available data and the lack of reporting on our outcomes of interest might limit the results and conclusions of this systematic review."

Introduction:
Sentence 3 would be easier to follow if the mechanisms were numbered
We have modified the sentence as follows: HCM arises due to four main mechanisms: 1) humoral secretion of parathyroid hormone related peptide (PTHrP) accounts for over 80% of cases and occurs most commonly in breast cancer and squamous cell carcinoma of the lung, head and neck, and the kidney, 2) local osteolytic release of calcium, known as local osteolytic hypercalcemia (LOH), such as seen with multiple myeloma and some breast cancers, 3) high levels of calcitriol (1,25-dihydroxyvitamin D) such as noted in leukemias, HTLV1, and some lymphomas, or secretion of the native parathyroid hormone (PTH) from a carcinoma, or 4) ectopic PTH secretion by some cancers including neuroendocrine tumors (Table 1) [4, 5].

Line 25 – is this statistic up-to-date? It seems like this statistic comes from a small study (126 patients) published in 1990 and not reference 7. Reference 8 talks about a mortality rate in hospital of 6.8%. Please ensure accurate and up-to-date survival outcome estimates are cited.
We have referenced the original study as recommended (now reference 8) and deleted references 7 and 8. We were unable to identify a more recent study on survival.

RANKL needs to be spelled out – page 3, line 38
We have modified the sentence as follows: "In 2014, Denosumab a Receptor activator of nuclear factor kappa-B ligand (RANKL) inhibitor has been approved for the treatment of HCM refractory to bisphosphonates with significant efficacy"

GRADE should be spelled out here – page 3, line 55
We have modified the sentence as follows: The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) Working Group has developed an Evidence-to-Decision (EtD) framework for the assessment of factors that should complement evidence on the benefits and harms when guideline groups make recommendations.

Methods
What is the rationale for limiting to the last 10 years? Please include those reasons in the protocol. Why was there a search also done with no time limit and only in MEDLINE?
The concept and therefore literature regarding decisional frameworks is relatively new thus the limitation of the timeframe to 10 years. We have included an additional search without time limit to enhance coverage. We modified Methods as follows: "We will test and use these two search strategies with varying sensitivity and specificity developed by two independent people to obtain better coverage of the literature."

Consider adding percent agreement and kappa statistics for agreement between reviewers in each stage.
We added the following to the Study Selection Section under Methods: "We will measure the

agreement between the two reviewers (AB, MR) at each screening step using the Cohen's kappa statistic."

This review may benefit from also screening any studies that have cited your included studies – as part of the hand search. Please consider including in the protocol.
We have modified the last sentence in the Information sources and search strategy section as follows: "We will also try to identify papers by hand searching references from the included studies and studies that have cited the included studies."

For eligibility – please specify the types of observational studies included.
We have modified the sentence as follows: "We will include observational studies (cohort, cross-sectional and case-control studies), trials, reviews and qualitative studies conducted in adult patients (≥ 18 years of age) with hypercalcemia of malignancy."

Abbreviations should be footnoted in the tables
We added the abbreviations as footnotes under Tables 1 and 2.